A: TGS + explicit PD

B: TGS + PhysX joint drive

C: PGS + PhysX joint drive

rotor inertia $(10^{-3}$ kg m$^2)$

theory: slope 1.00
fit: slope 1.12, R^2 = 0.9367

theory: slope 1.00
fit: slope 3.59, R^2 = 0.6650

theory: slope 1.00
fit: slope -1.37, R^2 = 0.6748

$h = 1/f$ (ms)

ΔK_D $(10^{-2}$ N m s/rad)

theory: slope 4.00
fit: slope -1.12, R^2 = 0.6492

theory: slope 4.00
fit: slope 12.92, R^2 = 0.6441

theory: slope 4.00
fit: slope -8.16, R^2 = 0.9573

$h = 1/f$ (ms)

ΔK_P (N m/rad)

theory: slope 0.00
fit: quadratic, R^2 = 0.9851

theory: slope 0.00
fit: quadratic, R^2 = 0.8787

theory: slope 0.00
fit: quadratic, R^2 = 0.8733

$h = 1/f$ (ms)

post-fit RMSE (rad)

theory: RMSE $\propto h^2$

h^2 (ms^2)

Isaac Sim 5.1 / PhysX. Parameter rows: first-order theory (solid), fit (dashed). Last row: post-fit RMSE against h^2.